Retail Sales Analytics Project

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Role Context: Data Analytics Portfolio (Python & Power BI)

1. Executive Summary

This document outlines the architecture, methodologies, and technical implementations for the Retail Sales Analytics Capstone Project. The objective was to design a robust, end-to-end data pipeline capable of analyzing sales performance, customer behavior, product profitability, and temporal sales trends to drive informed business decisions.

The project successfully integrates a Python-based automated data processing backend hosted in Google Colab with an interactive, multi-page business intelligence dashboard developed in Power BI.

2. Technical Architecture & Data Pipeline

A key achievement of this project is the implementation of an automated, persistent data pipeline, shifting away from manual file uploads toward a cloud-integrated architecture.

2.1 Cloud Integration

- **Storage Solution:** Raw dataset files (`Sales_10000.xlsx` , `Customers.xlsx` , `Stores.xlsx` , `Products.xlsx`) are centrally hosted within a dedicated Google Drive repository (`Capstone_Data`).
- **Automated Ingestion:** The Python backend connects directly to the Google Drive API using `google.colab.drive` , enabling seamless, automated data extraction upon runtime initialization.

Backend Automation Logic (Python)

```
from google.colab import drive
drive.mount('/content/drive')

# Automated Data Ingestion
sales = pd.read_excel('/content/drive/MyDrive/Capstone_Data/Sales_10000.xlsx')
customers = pd.read_excel('/content/drive/MyDrive/Capstone_Data/Customers.xlsx')
```

3. Exploratory Data Analysis (EDA) - Python Backend

The project utilizes the Pandas library to perform deep-dive Exploratory Data Analysis (EDA). The scripting approach was intentionally selected over UI-based dashboard reporting for these specific queries to showcase strong programmatic logic and data manipulation capabilities.

Analytical Query	Python Implementation (Pandas)	Primary Finding
Highest Revenue City	`df.groupby('City')['Revenue'].sum()`	Bengaluru / Mumbai
Top Selling Category	`df.groupby('Category')['Quantity'].sum()`	Clothing
Popular Payment Method	`df['PaymentMethod'].value_counts()`	UPI
Customer Spending Segment	`df.groupby('AgeGroup')['Revenue'].sum()`	Analyzed via grouping logic

4. Power BI Dashboard Development

The presentation layer consists of a finalized, two-page Power BI dashboard designed for executive stakeholders. The design prioritizes visual clarity and avoids unnecessary clutter by maintaining strict boundaries between programmatic EDA reporting and visual KPI tracking.

4.1 Key Visualizations

- **Geospatial Revenue Mapping:** Utilizing map visuals to track high-performing locations.
- **Profitability Treemaps:** Highlighting store-level and brand-level profitability metrics.
- **Temporal Trend Lines:** Tracking monthly maximum sales peaks using date hierarchies.

4.2 Strategic Design Choice

The dashboard was strictly constrained to two pages to ensure a clean, professional aesthetic. Redundant visualizations for the 8 core EDA questions were explicitly omitted from the Power BI interface, as they are comprehensively addressed within the Python backend codebase, reflecting a balanced, full-stack approach to data analytics.

5. Conclusion

The Retail Sales Analytics Capstone represents a fully functional, automated data pipeline. By integrating Google Drive persistent storage, Pandas-driven analytics, and Power BI visualization, the project successfully answers critical business questions while demonstrating robust backend logic suitable for professional software and data engineering environments.